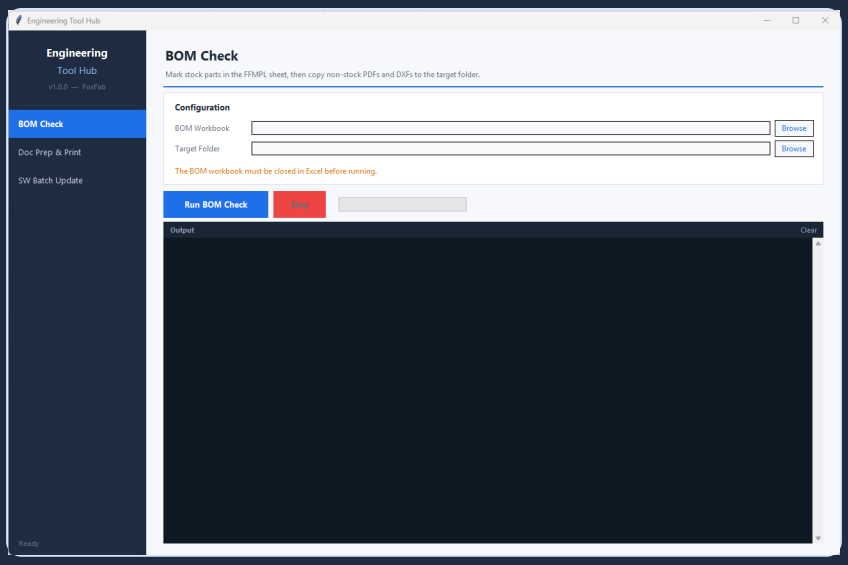
Engineering Tool Hub

User Guide

Version 1.0.0



Lambert Badong

Revision: Initial Release

FoxFab Internal Documentation

March 2026

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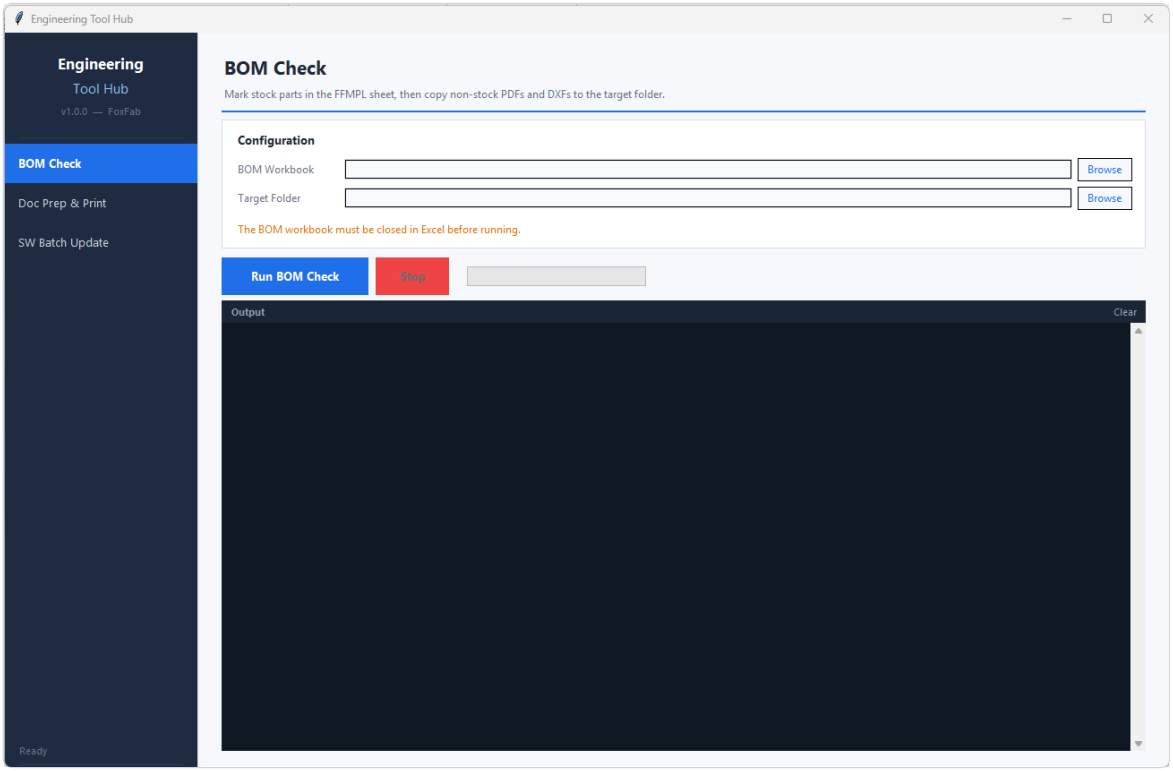
1 Introduction

Engineering Tool Hub is a FoxFab desktop application that combines three manufacturing workflow tools into a single interface. Instead of running separate scripts or macros, everything is accessible from one window with a consistent layout.

BOM Check	Marks stock parts and copies the correct-revision PDFs and DXFs for non-stock parts into the job folder.
Doc Prep & Print	Builds and prints a complete manufacturing packet — FWO, BOM, CNC files, flats, assemblies — in the correct order.
SW Batch Update	Guides you through running the SolidWorks macro to batch-update custom properties and export DXFs.

1.1 Application Layout



The window is divided into two main areas:

- **Sidebar (left)** — Click any tool name to switch to it instantly. The active tool is highlighted in blue.
- **Content panel (right)** — Shows the selected tool's config fields, action buttons, progress bar, and terminal output.

1.1.1 Terminal colour guide

Colour	Tag	Meaning
Blue	`[INFO]`	General progress update
Green	`[OK]`	Action completed successfully
Amber	`[WARN]`	Something to note, not an error
Red	`[ERROR]`	Something went wrong

2 Getting Started

2.1 Requirements

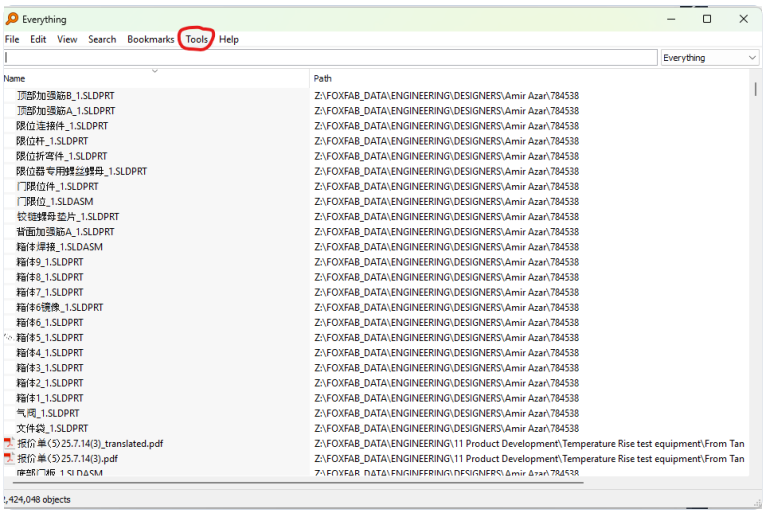
Make sure the following are available on your machine before using the app:

Requirement	Notes
Python (latest)	Download from python.org. During installation, tick **Add Python to PATH**.
Microsoft Excel	Must be installed. Used to read and write BOM spreadsheets.
Adobe Acrobat 2017+	Required for printing. Standard Reader is not sufficient.
Everything (search)	The Everything service must be running and have indexed the `Z:\` drive.
Network drive `Z:\`	Must be mapped and accessible.
FoxFab printer	`\\NPSVR05\FoxFab (Konica Bizhub C360i)` must be installed.

2.2 Configuring Everything HTTP Server

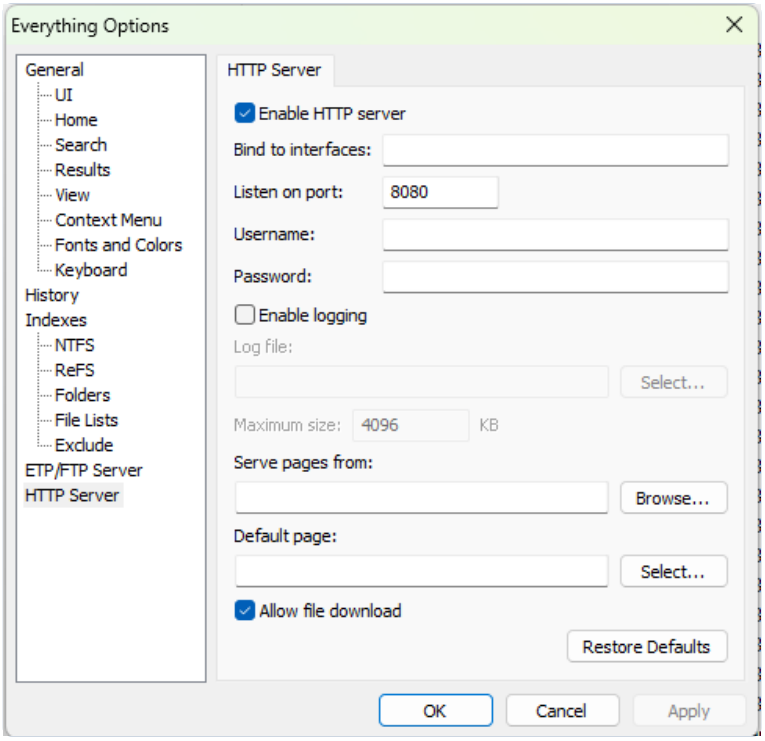
In addition to having the Everything service running, the HTTP server must be enabled for the application to query it.

Step 1. Open the **Everything** application.



Step 2. Go to **Tools** → **Options**.

Step 3. In the left sidebar, select **HTTP Server**.



Step 4. Tick **Enable HTTP server**.

Step 5. In the **Listen port** field, enter `8080`.

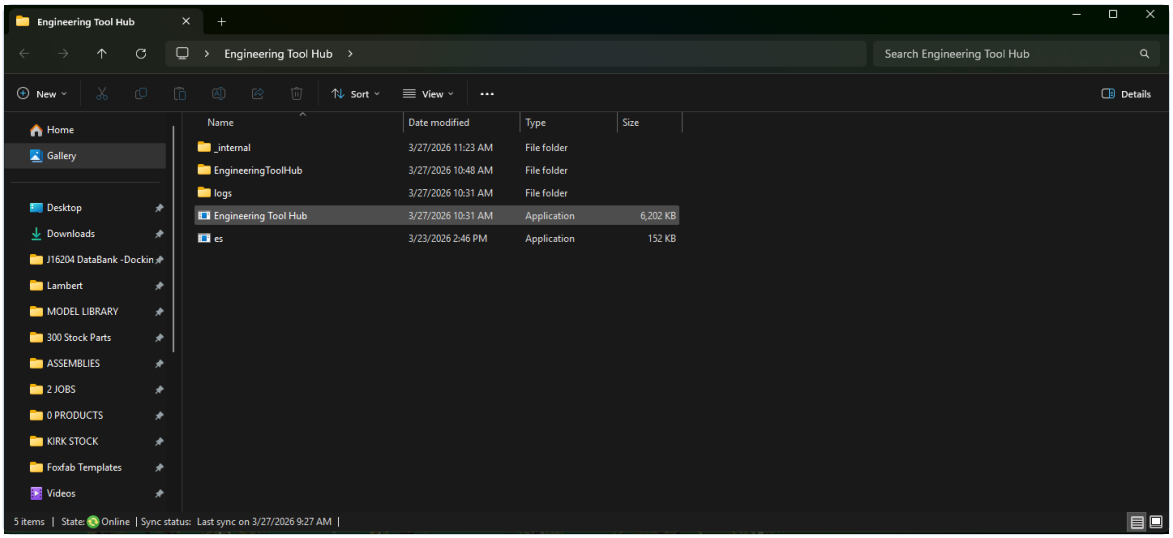
Step 6. Click **OK** to save and close the options window.

! *The HTTP server must be enabled and listening on port `8080` before running BOM Check. If the port is already in use by another application, the Everything search will fail.*

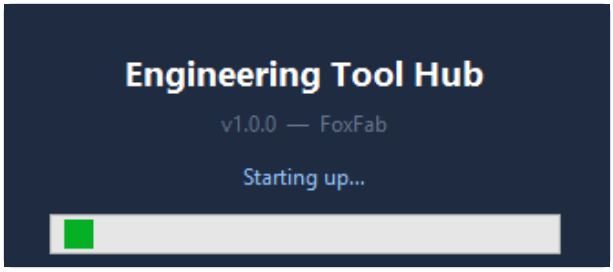
2.3 Launching the Application

Step 7. Navigate to the `Engineering Tool Hub` folder on the shared drive or your desktop.

Step 8. Double-click **Engineering Tool Hub.exe** to launch.

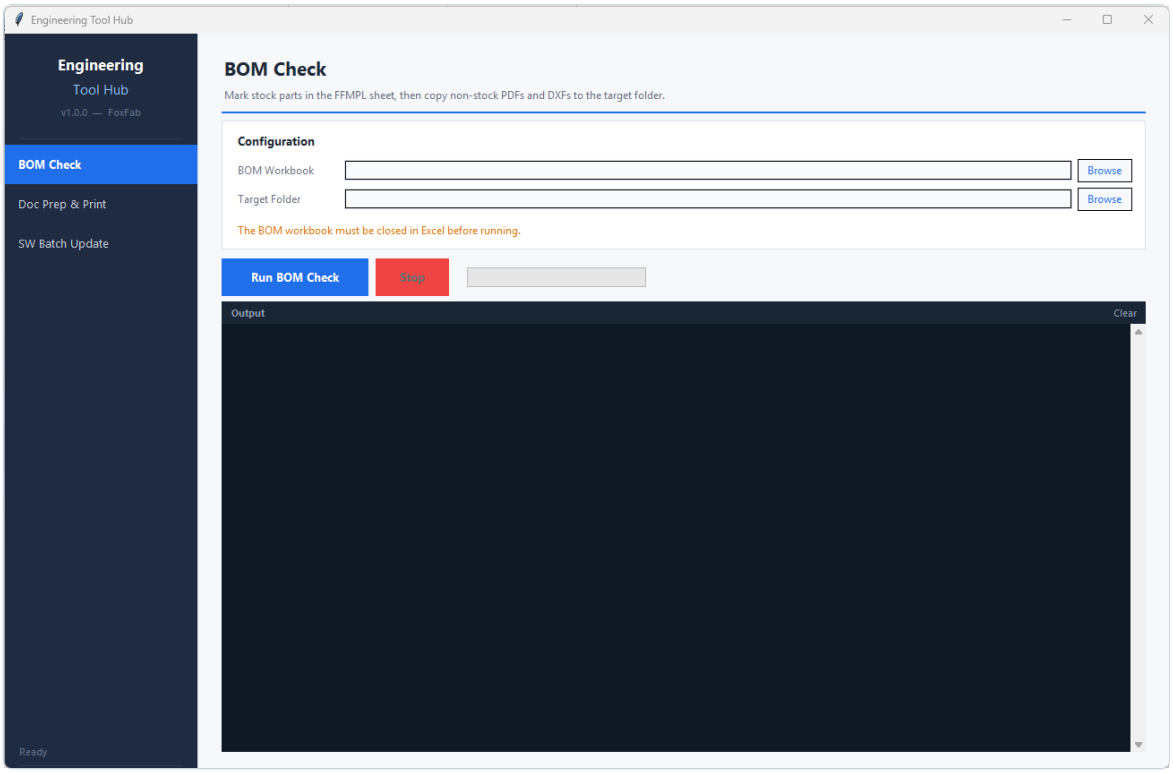


A loading screen appears briefly while the app starts.



> ! *The entire* `Engineering Tool Hub` *folder is required. Do* ***not*** *move just the* `.exe` *file — it will fail to start without the* `_internal` *folder alongside it.*

Step 9. The main window opens on the **BOM Check** panel by default.


Engineering Tool Hub
Engineering
Tool Hub
v1.0.0 — FoxFab
BOM Check
Doc Prep & Print
SW Batch Update
Ready
BOM Check
Mark stock parts in the FFMPL sheet, then copy non-stock PDFs and DXFs to the target folder.
Configuration
BOM Workbook
Browse
Target Folder
Browse
The BOM workbook must be closed in Excel before running.
Run BOM Check
Stop
Output
Clear

3 BOM Check

BOM Check reads the **FFMPL** sheet from a job BOM workbook and runs two passes:

1. **Pass 1 — Stock Parts Check.** Each part number is searched against the Stock Parts folder. Matching parts are marked in the spreadsheet.
2. **Pass 2 — Non-Stock File Copy.** For every non-stock part, the highest-revision PDF and DXF are found and copied to your target folder.

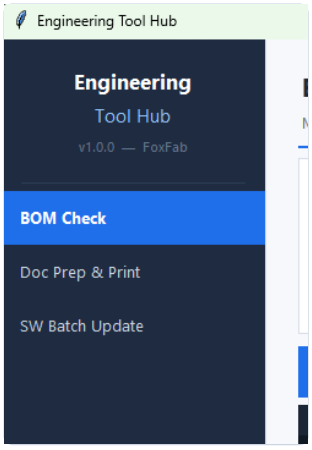
3.1 Before You Start

> ! *The BOM Excel file **must be closed** before running BOM Check. If it is open in Excel, the tool cannot write to it and will report an error.*

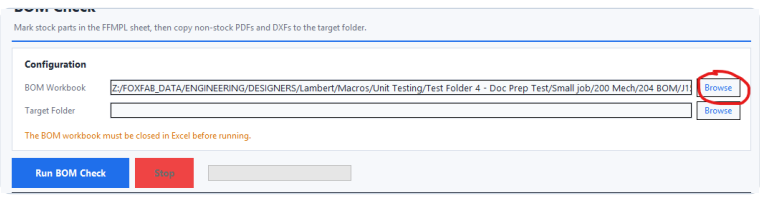
> ! *Make sure the **Everything** service is running. Open the Everything application and confirm the file count is not zero.*

3.2 Step-by-Step

Step 1. Click **BOM Check** in the sidebar.

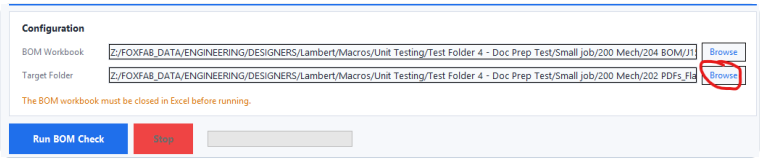


Step 2. In the **BOM Workbook** field, click **Browse** and select the job BOM `.xlsx` or `.xlsm` file. You can also type the path directly.



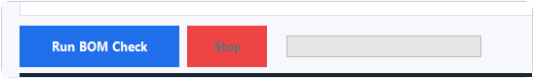
> **i** *If you browse to a folder instead of a file, the app will list all Excel files inside it and let you pick one.*

Step 3. In the **Target Folder** field, click **Browse** and select the folder where non-stock PDFs and DXFs should be copied. This is usually the job's `202 PDFs_Flats` folder.

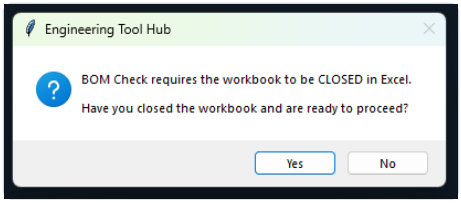


> **i** *If a `202 PDFs_Flats` folder is detected automatically next to the BOM, it will be offered as the default. You can accept or change it.*

Step 4. Close Excel (if open), then click **Run**.



Step 5. A confirmation prompt appears asking you to verify Excel is closed. Click **Yes** to proceed.

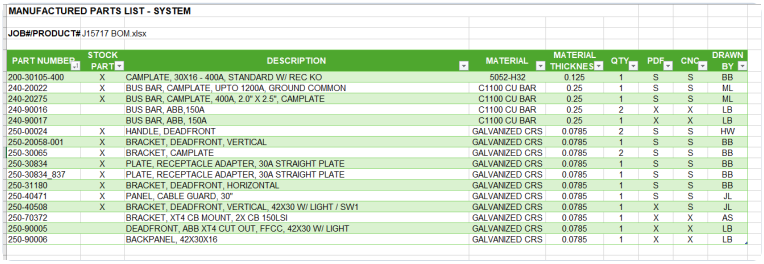


Step 6. Watch the terminal for progress. Pass 1 runs first, then Pass 2.

```
[INFO] Opening workbook:  Z:\Jobs\26-0042\204 BOM\BOM rA.xlsx

============================================================
PASS 1 -- Stock Parts Check
Searching:  Z:\...300 Stock Parts\PDFs & Flats
============================================================
Checking:  300-1005      FOUND -- marked X / S / S
Checking:  240-2201      not found
Checking:  240-2202      not found

Pass 1 complete:  1 of 3 parts found in stock.

============================================================
PASS 2 -- Non-Stock PDF + DXF Copy
============================================================
```

```
240-2201  rB  PDF:COPIED  DXF:COPIED
240-2202  rC  PDF:EXISTS  DXF:EXISTS

[OK] Workbook saved.  Pass 2 complete.
```

Step 7. Once complete, the progress bar fills and a summary is shown. Open the BOM in Excel to verify the markings.



MANUFACTURED PARTS LIST - SYSTEM

JOB#/PRODUCT# J15717 BOM.xlsx

PART NUMBER	STOCK PART	DESCRIPTION	MATERIAL	MATERIAL THICKNES	QTY	PDF	CNC	DRAWN BY
200-30105-400	X	CAMPLATE, 30X16 - 400A, STANDARD W/ REC KO	5052-H32	0.125	1	S	S	BB
240-20022	X	BUS BAR, CAMPLATE, UPTO 1200A, GROUND COMMON	C1100 CU BAR	0.25	1	S	S	ML
240-20275	X	BUS BAR, CAMPLATE, 400A, 2.0" X 2.5", CAMPLATE	C1100 CU BAR	0.25	1	S	S	ML
240-90016		BUS BAR, ABB,150A	C1100 CU BAR	0.25	2	X	X	LB
240-90017		BUS BAR, ABB, 150A	C1100 CU BAR	0.25	1	X	X	LB
250-00024	X	HANDLE, DEADFRONT	GALVANIZED CRS	0.0785	2	S	S	HW
250-20058.001	X	BRACKET, DEADFRONT, VERTICAL	GALVANIZED CRS	0.0785	1	S	S	BB
250-30065	X	BRACKET, CAMPLATE	GALVANIZED CRS	0.0785	2	S	S	BB
250-30834	X	PLATE, RECEPTACLE ADAPTER, 30A STRAIGHT PLATE	GALVANIZED CRS	0.0785	1	S	S	BB
250-30834_837	X	PLATE, RECEPTACLE ADAPTER, 30A STRAIGHT PLATE	GALVANIZED CRS	0.0785	1	S	S	BB
250-31180	X	BRACKET, DEADFRONT, HORIZONTAL	GALVANIZED CRS	0.0785	1	S	S	BB
250-40471	X	PANEL, CABLE GUARD, 30"	GALVANIZED CRS	0.0785	1	S	S	JL
250-40508	X	BRACKET, DEADFRONT, VERTICAL, 42X30 W/ LIGHT / SW1	GALVANIZED CRS	0.0785	1	X	S	JL
250-70372		BRACKET, XT4 CB MOUNT, 2X CB 150LSI	GALVANIZED CRS	0.0785	1	X	X	AS
250-90005		DEADFRONT, ABB XT4 CUT OUT, FFCC, 42X30 W/ LIGHT	GALVANIZED CRS	0.0785	1	X	X	LB
250-90006		BACKPANEL, 42X30X16	GALVANIZED CRS	0.0785	1	X	X	LB

3.3 Understanding the Results

Column	Value	Meaning
B — STOCK PART	`X`	Part found in the stock library
G — PDF	`S`	Stock part — file sourced from stock
G — PDF	`X`	Non-stock — PDF/DXF copied to target folder
H — CNC	`S`	Stock part — CNC sourced from stock

3.4 Stopping Early

Click **Stop** at any time to halt the run after the current part finishes. The workbook will not be saved if stopped mid-pass — re-run from the beginning.

4 Doc Prep & Print

Doc Prep & Print builds a complete manufacturing packet from a job folder and sends it to the printer in the correct order, or saves it locally in Simulation Mode.

4.1 Overview — What Gets Printed

#	Document	Duplex	Notes
01	Fabrication Work Order	Simplex	Auto-filled from PRF data
02	BOM (all sheets)	Simplex	CNC column H already marked
03–N	CNC files (duplex)	Duplex	One PDF per file
N+1	CNC Simplex (merged)	Simplex	All simplex CNCs merged
N+2	PDFs_Flats (merged)	Simplex	All flat PDFs merged
N+3	Production Release Form	Simplex	First sheet only
N+4	Electrical Pack	Simplex	Pages 1–2 only
N+5	Assemblies (merged)	Simplex	All assembly PDFs merged

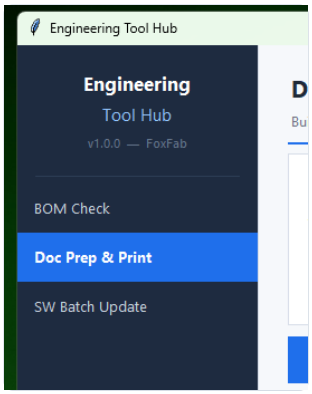
4.2 Before You Start

! *Close the BOM Excel file before clicking* ***Generate Documents****. The CNC column marking step opens the BOM via Excel — it will fail if the file is already open.*

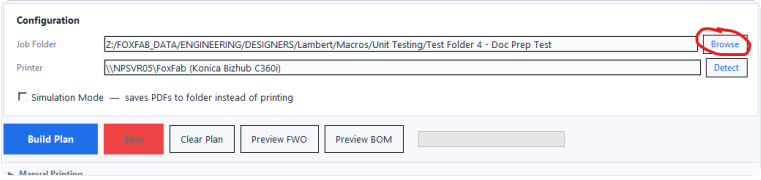
! *Adobe Acrobat (not Reader) must be installed for COM printing. If it is not found, the app will fall back to a slower printing method.*

4.3 Step-by-Step

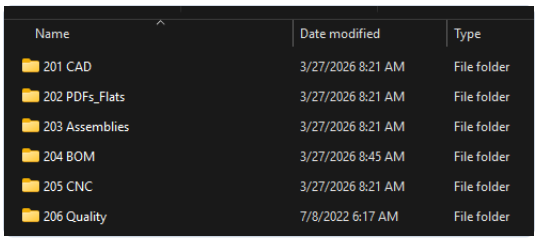
Step 1. Click **Doc Prep & Print** in the sidebar.



Step 2. In the **Job Folder** field, click **Browse** and select the job's **variant folder** — the one that contains `204 BOM\`, `205 CNC\`, and `202 PDFs_Flats\` inside it.



> *The variant folder is usually named something like `J26-0042-01` or `01 Standard`. Select the folder* containing *the `204 BOM` folder, not the `204 BOM` folder itself.*



Step 3. Check that the **Printer** field shows the correct printer name. The default is the FoxFab Konica Bizhub. Change it only if printing elsewhere.

Step 4. Click **Build Plan**.

Step 5. The terminal scans the job folder and lists everything it found. Review the output to confirm all expected documents are detected.

```
[INFO] Building plan for:  Z:\Jobs\26-0042\01 Standard
[OK] BOM: BOM rA.xlsx
[OK] PRF: PRF J26-0042.xlsx
[OK] FWO template:  FWO_Template.pdf
[OK] CNC duplex:  3 files
[OK] CNC simplex:  5 files
[OK] PDFs_Flats:  8 files
[WARN] Electrical pack not found -- will be skipped
[OK] Plan ready.  Click Generate Documents to continue.
```

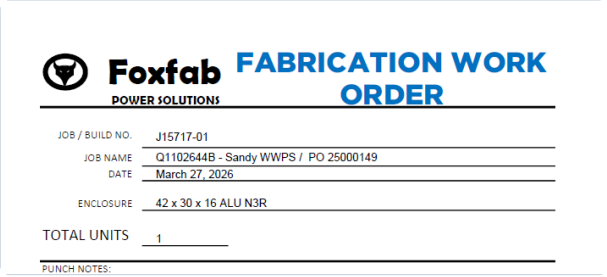
> *Items shown as `[WARN]` will be skipped — the run does not stop on a missing document.*

4.4 Preview Options (Optional)

Before committing to a full print run, use the preview buttons to verify key documents look correct.

4.4.1 Preview FWO

Click **Preview FWO** to fill and open the Fabrication Work Order without printing it. The result opens automatically from `logs\FWO_preview.pdf`.



Foxfab POWER SOLUTIONS

FABRICATION WORK ORDER

JOB / BUILD NO. J15717-01

JOB NAME Q1102644B - Sandy WWPS / PO 25000149

DATE March 27, 2026

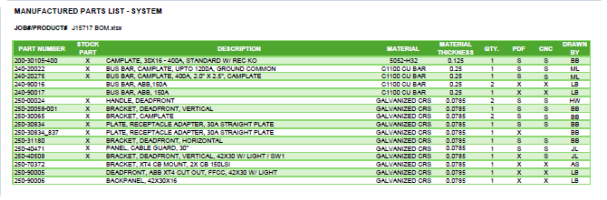
ENCLOSURE 42 x 30 x 16 ALU N3R

TOTAL UNITS 1

PUNCH NOTES:

4.4.2 Preview BOM

Click **Preview BOM** to run the CNC marking pass first, then export and open the BOM as a PDF. Use this to confirm column H is correct before printing.



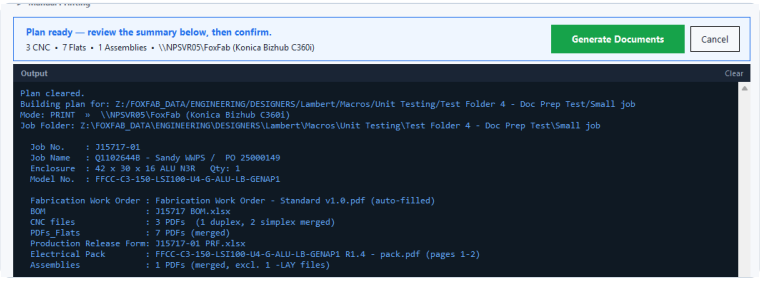
MANUFACTURED PARTS LIST - SYSTEM

JOB#/PRODUCT# J15717 BOM.xlsx

PART NUMBER	STOCK PART	DESCRIPTION	MATERIAL	MATERIAL THICKNESS	QTY.	PDF	CNC	DRAWN BY
200-30105-400	X	CAMPLATE, 30X16 - 400A, STANDARD W/ REC KO	5052-H32	0.125	1	S	S	BB
240-20022	X	BUS BAR, CAMPLATE, UPTO 1200A, GROUND COMMON	C1100 CU BAR	0.25	1	S	S	ML
240-20278	X	BUS BAR, CAMPLATE, 400A, 2.0" X 2.5", CAMPLATE	C1100 CU BAR	0.25	1	S	S	ML
240-90116		BUS BAR, ABB, 160A	C1100 CU BAR	0.25	2	X	X	LB
240-90117		BUS BAR, ABB, 160A	C1100 CU BAR	0.25	1	X	X	LB
250-00024	X	HANDLE, DEADFRONT	GALVANIZED CRS	0.0795	2	S	S	HW
250-20066-001	X	BRACKET, DEADFRONT, VERTICAL	GALVANIZED CRS	0.0795	1	S	S	BB
250-30064	X	BRACKET, CAMPLATE	GALVANIZED CRS	0.0795	2	S	S	BB
250-30834	X	PLATE, RECEPTACLE ADAPTER, 30A STRAIGHT PLATE	GALVANIZED CRS	0.0795	1	S	S	BB
250-30834_837	X	PLATE, RECEPTACLE ADAPTER, 30A STRAIGHT PLATE	GALVANIZED CRS	0.0795	1	X		BB
250-31180	X	BRACKET, DEADFRONT, HORIZONTAL	GALVANIZED CRS	0.0795	1	S	S	BB
250-40471	X	PANEL, CABLE GUARD, 30"	GALVANIZED CRS	0.0795	1	S	S	JL
250-40908	X	BRACKET, DEADFRONT, VERTICAL, 42X30 W/ LIGHT / BW1	GALVANIZED CRS	0.0795	1	X	S	JL
250-70372		BRACKET, XT4 CB MOUNT, 24 CB 160LSI	GALVANIZED CRS	0.0795	1	X	X	AG
250-90005		DEADFRONT, ABB XT4 CUT OUT, FFCC, 42X30 W/ LIGHT	GALVANIZED CRS	0.0795	1	X	X	LB
250-90006		BACKPANEL, 42X30X16	GALVANIZED CRS	0.0795	1	X	X	LB

> ! *Preview BOM opens the BOM via Excel COM. Close Excel before clicking it.*

> **Step 6.** Once the plan looks correct, click **Generate Documents**.



The app will automatically:

a. Scan `205 CNC\` and mark column H of the BOM with `X` for every matched part, then save the workbook.

b. Generate all documents into a temporary folder in print order.

c. Auto-populate the Manual Printing list with every generated PDF.

```
[INFO] CNC Column Marker:  scanning 205 CNC...
[OK] Found 8 CNC PDFs in 205 CNC
[OK] Resolved 14 unique part numbers
[OK] BOM marked and saved.
[INFO] Generating 01_FWO...
[OK] 01_FWO_J26-0042.pdf
[INFO] Generating 02_BOM...
[OK] 02_BOM_rA.pdf
...
[OK] All documents generated.  Ready to print.
```

Step 7. An **amber confirmation bar** appears listing every document queued for printing. Read through the list before proceeding.

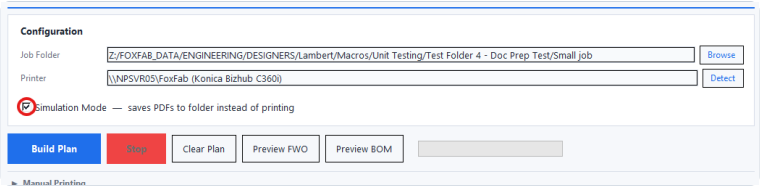
Step 8. Click **Send to Printer** when ready. All documents are sent to Acrobat in order. Do not close the app while printing is in progress.

```
[INFO] Starting Acrobat COM session...
[OK] Printing 01_FWO_J26-0042.pdf   (simplex)
[OK] Printing 02_BOM_rA.pdf       (simplex)
[OK] Printing 03_CNC_J26042-01.pdf  (duplex)
...
[OK] All documents sent.  Acrobat session closed.
```

ℹ *Clicking **Cancel Print** instead sends nothing to the printer and clears the queue. You can adjust anything and click Generate Documents again.*

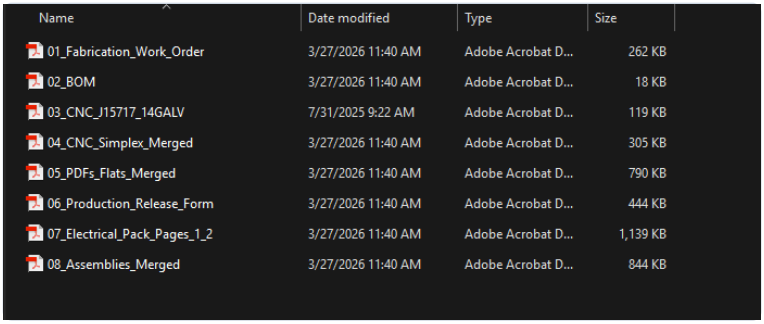
4.5 Simulation Mode

Tick the **Simulation Mode** checkbox before clicking **Send to Printer** to save all documents locally instead of printing. This is useful for reviewing output on a machine without a printer.



Documents are saved to a timestamped folder next to the application:

`Simulated_Print_Output_J26-0042_20260327_091500\`



Name	Date modified	Type	Size
01_Fabrication_Work_Order	3/27/2026 11:40 AM	Adobe Acrobat D...	262 KB
02_BOM	3/27/2026 11:40 AM	Adobe Acrobat D...	18 KB
03_CNC_J15717_14GALV	7/31/2025 9:22 AM	Adobe Acrobat D...	119 KB
04_CNC_Simplex_Merged	3/27/2026 11:40 AM	Adobe Acrobat D...	305 KB
05_PDFs_Flats_Merged	3/27/2026 11:40 AM	Adobe Acrobat D...	790 KB
06_Production_Release_Form	3/27/2026 11:40 AM	Adobe Acrobat D...	444 KB
07_Electrical_Pack_Pages_1_2	3/27/2026 11:40 AM	Adobe Acrobat D...	1,139 KB
08_Assemblies_Merged	3/27/2026 11:40 AM	Adobe Acrobat D...	844 KB

4.6 Manual Printing

At the bottom of the Doc Prep panel is a collapsible » **Manual Printing** section. Click the arrow to expand it.

- After **Generate Documents** the list is **auto-populated** with every generated PDF in print order, duplex pre-set per file.
- Click **Add File** to add any extra PDF manually.
- Click **X** next to a file to remove it from the list.
- **Print** sends the selected file immediately (background thread — UI stays responsive).
- **Print All** sends all files in list order through the same Acrobat session.

Use Manual Printing to resend a single document or the full batch without re-running the entire plan.

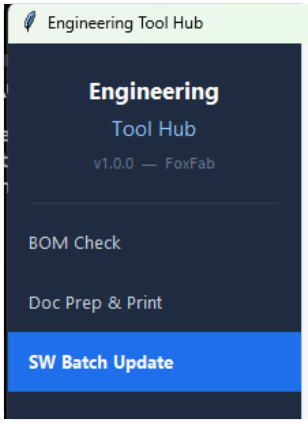
5 SW Batch Update

SW Batch Update guides you through running the SolidWorks VB macro that batch-updates custom properties and exports DXF flat patterns.

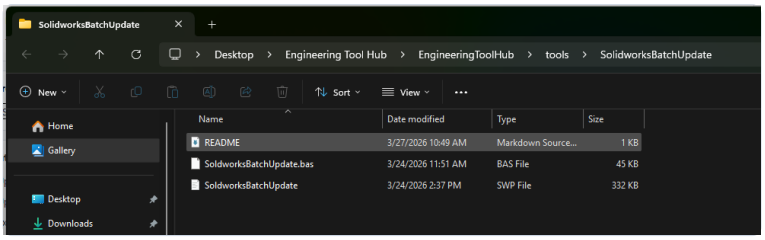
> ! *SolidWorks must be installed on your machine. This tool cannot function without it.*

5.1 Step-by-Step

Step 1. Click **SW Batch Update** in the sidebar.

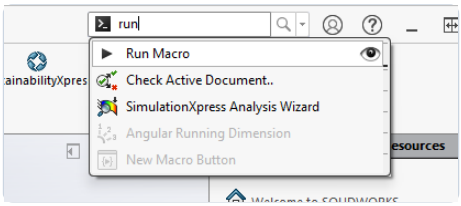


Step 2. The panel shows the path to the macro file. Click **Open Folder** to open the containing folder in File Explorer.



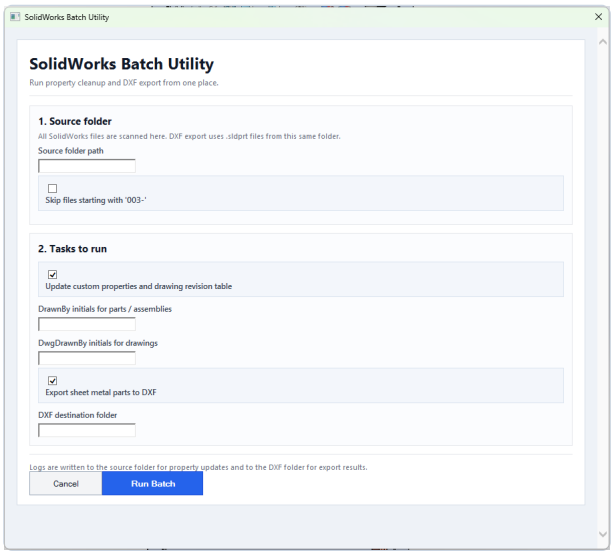
Step 3. Open SolidWorks.

Step 4. In SolidWorks, go to **Tools** → **Macro** → **Run**.



Step 5. Browse to `SoldworksBatchUpdate.bas` and click **Open**.

Step 6. The macro dialog opens. Enter your initials, select the source folder and output folder, then click **Run**.



Step 7. The macro updates `DrawnBy` and `DwgDrawnBy` properties across all parts in the folder and exports flat-pattern DXF files.

6 Troubleshooting

6.1 The application will not start

- Ensure the entire `Engineering Tool Hub\` folder is present, not just the `.exe`.
- If files are missing from `_internal\`, re-run `build.bat` on the source machine and redistribute the whole folder.

6.2 BOM Check — `es.exe not found`

Everything CLI is missing. Check that `es.exe` is in the same folder as the `.exe` file.

6.3 BOM Check — "Could not open workbook"

The BOM file is still open in Excel. Close it fully (check the taskbar) and run again.

6.4 BOM Check — Pass 1 marks everything as "not found"

The Everything service is not running or has not indexed `Z:\`. Open the Everything application, go to **Tools → Rebuild Database**, and wait for indexing to complete before retrying.

6.5 Doc Prep — CNC folder not found automatically

The app looks for a folder beginning with `205` inside the job variant folder. If named differently, a folder picker will open — browse to the correct CNC folder and click **Select**. To skip CNC marking entirely, cancel the picker.

6.6 Doc Prep — FWO text in the wrong position

Open `app.py` in a text editor, find the `FWO_*` constants near the top, and adjust the `X/Y` values. Use **Preview FWO** after each change to verify without a print run.

6.7 Doc Prep — Acrobat not found

Adobe Acrobat (not Reader) must be installed. If installed in a non-standard location, add its full path to `ACROBAT_SEARCH_PATHS` in `app.py`.

6.8 Printing stops partway through

Use the **Manual Printing** panel to resend any documents that did not print. Add the missing files individually and click **Print** or **Print All**.

7 Quick Reference

7.1 Button Reference

Button / Element	Purpose
Sidebar tool name	Switch to that tool panel instantly
Build Plan	Scan the job folder and report what was found
Generate Documents	Mark CNC column, generate all PDFs in print order
Run	Start the BOM Check workflow
Stop	Halt the running task after the current step
Browse	Open a file or folder picker
Preview FWO	Fill and open the FWO PDF without printing
Preview BOM	Mark CNC column and open BOM PDF without printing
Send to Printer	Commit the document queue to the printer
Simulation Mode checkbox	Save output locally instead of printing
» Manual Printing	Expand/collapse the manual print queue

7.2 BOM Column Marking Reference

Column	Value	Meaning
B — STOCK PART	`X`	Found in stock library
G — PDF	`S`	Stock — file from stock folder
G — PDF	`X`	Non-stock — PDF/DXF copied to target
H — CNC	`S`	Stock — CNC from stock folder
H — CNC	`X`	CNC file found and matched in 205 CNC

7.3 Print Document Order

#	Document	Duplex	Pages
01	Fabrication Work Order	Simplex	All
02	BOM (all sheets)	Simplex	All
03–N	CNC duplex files	Duplex	All
N+1	CNC simplex (merged)	Simplex	All
N+2	PDFs_Flats (merged)	Simplex	All
N+3	Production Release Form	Simplex	1 only
N+4	Electrical Pack	Simplex	1–2 only
N+5	Assemblies (merged)	Simplex	All